

Video Transcript

Video 5: Creating *Containers* Using Python (03:31)

Creating containers by hand, they're certainly powerful. It gives us great speed and capabilities. This is what we have used to be able to bypass the local configuration and set up that can be quite lengthy. However, when we're creating a lot of these, doing it by hand, one by one, at the command line can be slow. And so it would be much better to have a Python program that ran all of those commands against the command line on the shell for us. So, let's take a look and see how we can do that. As you can see here, I'm in my editor where I have a single file called `Pythos.py` and we'll start off here by adding two packages, the operating system and system package. Then we'll define an input argument, which in this case is called `create`. After that will come the function that we will use to create our containers. Note that it takes two parameters `command` in the database, and you can see here that in line 11, we're going to be creating that container, and if that goes well, we're going to use the name of the database simply to update the command line and know that we have successfully created that container.

Then we're going to use our block there, lines 15 through 18 to be able to read that input argument. And after that we will have a conditional statement to be able to go into that block of code to be able to create the container. As you can see here in this case, I'm going to be creating a container Docker run of my SQL and the rest of it are the parameters that we have used before. So, let's go ahead and save. I've gone ahead and moved to the desktop and I have opened a terminal, and if I request the path here or show you the path, you can see there that I am in the same folder. If I list the files, you can see there that I have containers and I've also opened my Docker dashboard to be able to see the container when it gets created. So, now let's go ahead and run our program, Python3, and the name of the file is `containers.py`. And then we're going to give that argument that we defined.

So, let's go ahead and hit return. And as you can see there that is pulling the container as we had expected and that is going through all of the setup and configuration that is needed. And you can see there that the dashboard just updated and I have a running instance of MySQL running on port 3306. And you can see there that it's nice and green, which means it's now

available and ready for me to use it. So, as you just saw, I went ahead and created the container programmatically for MySQL. It is now your turn. First, make sure you can replicate what I did, that you can create the MySQL container. Then use the command line instructions that we have used for the other databases, the Mongo database, Redis and Cassandra. And make sure you can add those instructions to that program and that you can create all of those containers in one step. It will make you feel powerful and it'll simply facilitate you being able to create all of these in one shot.

